ArtifactSweep

ahmedlamidi

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1 Introduction

Table 1: Synthetic degradation models and real-world evaluation data.

Degradation	**Forward model**	**Real-world data**
Low-dose	$y_i \sim \text{Poisson}\big(\alpha b_0\, e^{-[Ax]_i}\big) + \mathcal{N}(0, \sigma_e^2)$	Mayo LDCT-PD; AAPM–Mayo 2016; 2DeteCT
Sparse-view	$\tilde{p} = \mathcal{P}_\Omega(Ax)$ $\Omega = \{\theta_{k\Delta}\}_{k=0}^{N_v-1} \subset \Theta$	Subsampled raw: Mayo LDCT-PD, Walnut CBCT, 2DeteCT
Limited-angle	$\tilde{p} = \mathcal{P}_\Omega(Ax)$ $\Omega = [\theta_0,\ \theta_0 + \theta_{\max}]$	Cropped raw: Mayo LDCT-PD, Walnut CBCT, 2DeteCT
Beam hardening	$y_i = \int \eta(E)\, e^{-a_i^\top \mu(E)}\, dE$ $\tilde{p} = -\ln\big(y / \int \eta\big) \approx \sum_k c_k\, p^k$	2DeteCT (beam-hardening mode)
Metal / starvation	$\mu(E) = (1 - M) \odot \mu_t(E) + M \odot \mu_m(E)$ $\bar{y}_i = \int \eta(E)\, e^{-a_i^\top \mu(E)}\, dE$ $y_i \sim \text{Poisson}(\bar{y}_i + s_i) + \mathcal{N}(0, \sigma_e^2)$	CLINIC-metal; SpineWeb (no GT); AAPM CT-MAR 2023 (simulated)
Motion	$\tilde{p}(\theta) = A_\theta\big(\mathcal{T}_{\phi(\theta)}\, x\big)$	No paired benchmark; CQ500 (no GT)
Ring / detector	$\tilde{p}_{\theta,d} = g_d\, p_{\theta,d} + o_d$ $g_d = 0 \quad \forall d \in \mathcal{D}_{\text{dead}}$	TomoBank (real rings)

Notation and parameters. x is the clean attenuation image, A the system matrix with rows a_i, and $p = Ax$ the clean line integrals; degraded sinograms are obtained as $\tilde{p} = -\ln(y/b)$, where b is the unattenuated flux of each ray. In the low-dose model, b_0 is the blank-scan photon count per detector bin, $\alpha \in (0, 1]$ the dose fraction relative to full dose, and σ_e the standard deviation of electronic readout noise. For sparse-view and limited-angle acquisitions, Θ is the full angular set and $\mathcal{P}_\Omega$ keeps only the views in Ω: sparse-view keeps N_v evenly spaced views with step $\Delta = |\Theta|/N_v$ (typically 18–90 views), whereas limited-angle keeps a contiguous arc of length $\theta_{\max}$ starting at θ_0, with $\theta_{\max} < \pi$ for parallel beam and $\theta_{\max} < \pi + 2\gamma$ for a fan beam of half-angle γ. Beam hardening replaces the monoenergetic model with an integral over the source spectrum $\eta(E)$, set by tube voltage and filtration, and energy-dependent attenuation $\mu(E)$; the polynomial with coefficients c_k is the common single-material approximation. For metal, M is the binary metal mask, $\mu_t(E)$ and $\mu_m(E)$ the tissue and metal attenuation, $\bar{y}_i$ the expected polychromatic count, and s_i additive scatter; photon starvation arises when $b_0 e^{-a_i^\top \mu}$ approaches zero along rays through metal, and the clipping applied before the logarithm must be reported. For motion, A_θ is the projector at view θ and $\mathcal{T}_{\phi(\theta)}$ warps the object with view-dependent parameters $\phi(\theta)$: a rigid rotation and translation (R_θ, t_θ) or a deformable field. For ring artifacts, d indexes detector elements, $g_d \sim 1 + \mathcal{N}(0, \sigma_g^2)$ is the per-element gain error, o_d the offset error, and $\mathcal{D}_{\text{dead}}$ the set of dead elements.